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National Institute of Technology Goa

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, September 2026

Course Name: **Mechanics of Solids**

Date: **22/09/2026**

Duration : **90 minutes**

Course Code: **ME200**

Time: **10:00 AM - 11:30 AM**

Max. Marks: **50**

Assume missing data

1. A deformable component is subjected to an external loading and is immersed in a static fluid. In the reference configuration, a material particle is identified by $\underline{\mathbf{X}}$, and after deformation its position is $\underline{\mathbf{x}}(\underline{\mathbf{X}}, t)$. Using the concepts of traction and stress studied in class, answer the following:
 - (i) At the current position $\underline{\mathbf{x}}(\underline{\mathbf{X}}, t)$, a surface with unit normal $\underline{\mathbf{n}}$ is considered. Write the mathematical expression for the traction vector acting on this surface.
 - (ii) Using a suitable limiting mathematical expression, define the stress tensor at $\underline{\mathbf{x}}$. Clearly state one difference between the traction vector and the stress tensor at a point.
 - (iii) The component is now placed in a static fluid. What is the state of stress at any point inside the component due to the fluid pressure? Comment on the relationship between the traction vector and the surface normal $\underline{\mathbf{n}}$. [1+2+2]
2. A research group is developing mechanical models for different engineering materials. For a linearly elastic deformable body, the constitutive relation is given by $\underline{\underline{\sigma}} = \underline{\underline{\mathbf{C}}} \underline{\underline{\epsilon}}$, where $\underline{\underline{\mathbf{C}}}$ is a fourth-order stiffness tensor. In the most general case, $\underline{\underline{\mathbf{C}}}$ contains 81 components. However, by invoking the minor and major symmetries of the stiffness tensor, the number of independent elastic constants reduces to 21. The researchers want to characterize the elastic response of different materials through suitable laboratory experiments before developing their computational models. Based on the above, answer the following:
 - (i) Mild steel is commonly assumed to behave as an isotropic material. How many independent elastic constants are required to completely describe its linear elastic behaviour? Suggest suitable laboratory experiments that can be performed to determine these constants.
 - (ii) A wooden structural component exhibits different mechanical responses along its longitudinal, radial and tangential directions because of its internal structure. If wood is modeled as an orthotropic material, how many independent elastic constants are required? Briefly state the type of tests that could be used to characterize them.
 - (iii) The wall of a blood vessel has one preferred material direction associated with its fibre architecture. If the blood vessel is modeled as a transversely isotropic material, how many independent elastic constants are required? [2+2+1]

3. A mechanical component is subjected to a three-dimensional state of stress at a point $\underline{x}$. In the laboratory coordinate system $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$, the stress tensor is measured as

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix}.$$

For analysis of possible failure, the engineer transforms the coordinate system to $\{\underline{E}_1, \underline{E}_2, \underline{E}_3\}$, in which the stress tensor becomes diagonal:

$$\underline{\underline{\sigma}} = \begin{bmatrix} \hat{\sigma}_{xx} & 0 & 0 \\ 0 & \hat{\sigma}_{yy} & 0 \\ 0 & 0 & \hat{\sigma}_{zz} \end{bmatrix}.$$

- (i) What are the directions $\{\underline{E}_1, \underline{E}_2, \underline{E}_3\}$ called, and what is the physical significance of choosing these directions?
(ii) For the measured stress state

$$\sigma_{xx} = 0, \quad \sigma_{yy} = 2, \quad \sigma_{zz} = 6, \quad \tau_{xy} = 4, \quad \tau_{xz} = 6, \quad \tau_{yz} = 8,$$

determine $\hat{\sigma}_{xx}$, $\hat{\sigma}_{yy}$ and $\hat{\sigma}_{zz}$.

[1+4]

4. A critical point in a loaded mechanical component is subjected to the following three-dimensional stress state in the Cartesian basis $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$:

$$\underline{\underline{\sigma}}(\underline{x}) = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

- (i) Using Mohr's circle, determine the principal stresses and the maximum possible shear traction at the point. Clearly indicate the principal planes.
(ii) A plane passing through the point has its unit normal in the (xy) plane and is inclined at 7.5° clockwise from the x -axis. Determine the traction vector on this plane and resolve it into its normal and shear components.
(iii) Decompose the given stress tensor into its hydrostatic and deviatoric parts. Verify that the deviatoric stress tensor is traceless. [3+3+2]
5. A steel tie rod ($E = 2 \times 10^5 \text{ MN/m}^2$ and $\alpha = 12 \times 10^{-6}/^\circ\text{C}$) used to connect two rigid components in a machine frame is 3 cm in diameter and 5 m long. During operation, the rod is maintained at 95°C . The temperature subsequently drops to 30°C while the grips restraining the rod remain in place. Determine the thermal stress and the corresponding pull developed in the rod when
- (i) the grips remain elastic and do not yield, and
(ii) the grips yield by 0.12 cm due to the thermal contraction. [05]

An initially unstressed circular mark of diameter 200 mm is scribed on an 18 mm-thick aluminium plate, as shown in Figure 1. The plate is subsequently subjected to in-plane loading, producing normal stresses $\sigma_x = 85 \text{ MPa}$ and $\sigma_z = 150 \text{ MPa}$. For $E = 70 \text{ GPa}$ and $\nu = 1/3$, determine the change in:

- the lengths of diameters AB and CD
- the thickness and volume of the plate.

[08]

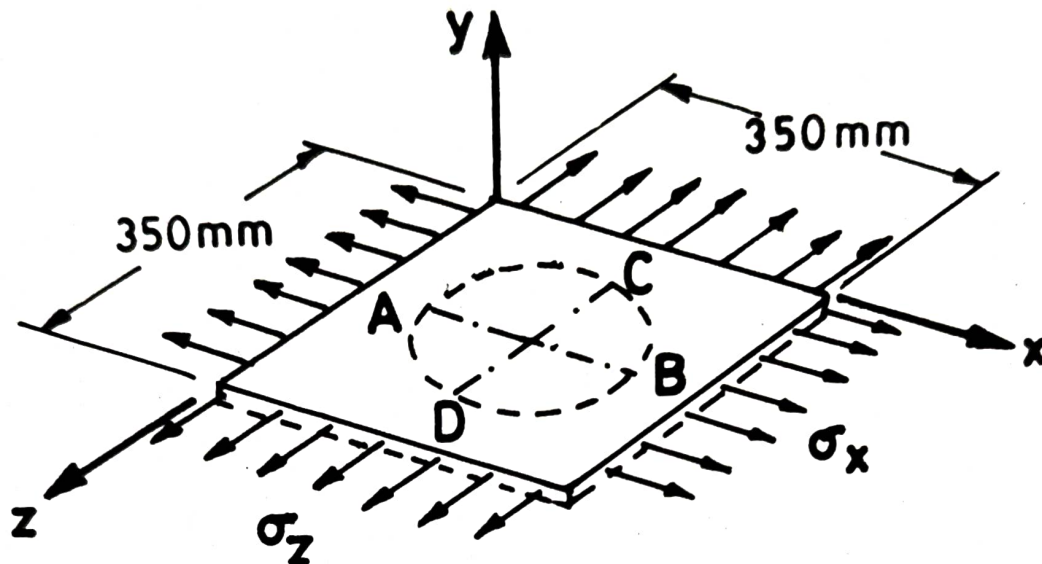


Figure 1

7. Determine the stresses in each wires supporting the rigid bar (see Figure 2) if $F = 20 \text{ kN}$. [07]

8. A composite bar (see Figure 3) is firmly attached to unyielding supports at the ends and is subjected to the axial load F . If the aluminum is stressed to 70 MPa , what is the stress in the steel? [07]

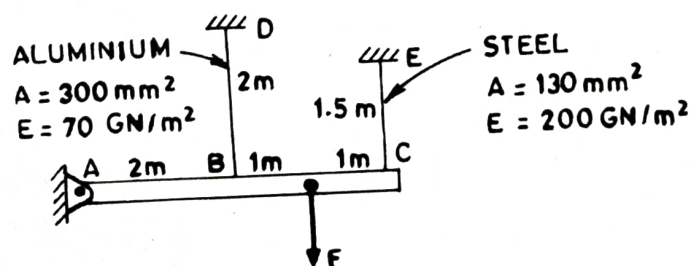


Figure 2

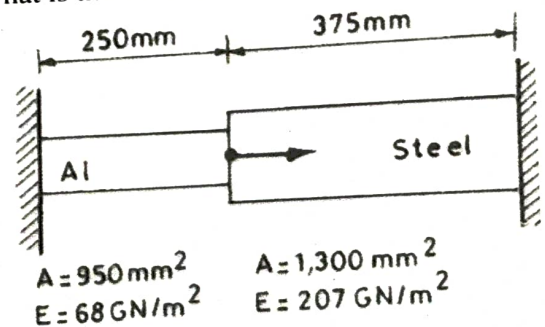


Figure 3

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